🔑 **Answer Key — Teacher Copy · Lesson 8-3**

MSTAR PRACTICE WORKSHEET

Lesson 8-3: Write and Solve Equations Using Multiplication or Division

Grade 6 Mathematics · Standard 6.AT.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Solve: 6x = 42

- **A** x = 7
- **B** x = 36
- **C** x = 48
- **D** x = 252

Divide both sides by 6: x = 42 ÷ 6 = 7. Check: 6 × 7 = 42 ✓

Question 2 SELECTED RESPONSE — CORRECT: D

Solve: n / 3 = 8

- **A** n = 3
- **B** n = 5
- **C** n = 11
- **D** n = 24

Multiply both sides by 3: n = 8 × 3 = 24. Check: 24 / 3 = 8 ✓

Question 3 SELECTED RESPONSE — CORRECT: A

Solve: 9m = 63

A m = 7

B m = 54

C m = 72

D m = 567

Divide both sides by 9: m = 63 ÷ 9 = 7. Check: 9 × 7 = 63 ✓

Question 4 SELECTED RESPONSE — CORRECT: D

Solve: y / 7 = 5

A y = 2

B y = 7

C y = 12

D y = 35

Multiply both sides by 7: y = 5 × 7 = 35. Check: 35 / 7 = 5 ✓

Question 5 SELECTED RESPONSE — CORRECT: A

A detective splits 54 clues equally among 9 case files. The equation 9c = 54 represents this. What is c?

A c = 6

B c = 9

C c = 45

D c = 63

Divide both sides by 9: c = 54 ÷ 9 = 6 clues per file.

Question 6 SELECTED RESPONSE — CORRECT: A

Vault A is sealed with 4x = 32. Crack the lock: what is x?

- **A** x = 8
- **B** x = 28
- **C** x = 36
- **D** x = 128

The 4 is multiplying x, so divide both sides by 4: x = 32 / 4 = 8. Check: 4 x 8 = 32.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

What is the solution of 6m = 54?

- **A** m = 9
- **B** m = 48
- **C** m = 60
- **D** m = 324

In 6m the coefficient 6 is multiplied by the variable, so divide both sides by 6: m = 54 ÷ 6 = 9. Check by substituting: 6 × 9 = 54.

- **B:** That subtracts 6 from 54. In 6m the 6 multiplies m, so divide instead.
- **C:** That adds 6 to 54. The 6 multiplies m, so divide both sides by it.
- **D:** That multiplies 6 by 54. To undo a multiplication, divide.

Part B — correct answer: A

Which reasoning justifies the step used to find that solution?

- **A** Division undoes multiplication, so dividing both sides by 6 isolates the variable.
- **B** Subtracting 6 from both sides removes the 6 that is written next to the variable.
- **C** Multiplying both sides by 6 isolates the variable because the same operation must be repeated.
- **D** The coefficient can simply be dropped because 6 is smaller than 54.

In 6m the variable is multiplied by 6, so the inverse operation is division. Dividing both sides by 6 keeps the equation balanced and gives m = 9. Subtracting 6 would give 48, and 6 × 48 is not 54.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, E

Select ALL equations that have x = 8 as their solution.

- **A** 4x = 32
- **B** x ÷ 2 = 4
- **C** 3x = 21
- **D** x ÷ 4 = 32
- **E** 56 = 7x

Substitute 8 for the variable in each equation. 4 × 8 = 32 is true, 8 ÷ 2 = 4 is true, and 56 = 7 × 8 is true. The third is false because 3 × 8 = 24, not 21; its solution is x = 7. The fourth is false because 8 ÷ 4 = 2, not 32; its solution is x = 128.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A bakery packed muffins into boxes of 8. They made 96 muffins total. Write an equation, solve it, and explain why your answer is reasonable. Then write a different situation that uses division and has the same answer.

Model answer: Eight muffins per box times an unknown number of boxes equals 96, so the equation is 8x = 96. Dividing both sides by 8 gives x = 12 boxes, and it is reasonable because 12 boxes × 8 muffins rebuilds all 96. A division situation with the same answer works too, such as 60 stickers shared equally by 5 students.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.